PDF ANONIMIZADO

ID publico: case_0225
Paginas del documento: 3

Copia anonimizada para verificacion metodologica.

Se removieron portada editorial, titulo, autores, afiliaciones, correos, DOI, URLs y metadatos del PDF original.
Las paginas restantes conservan el contenido util para revisar metodos, resultados, tablas y conclusiones.

Nota: la anonimización automatica prioriza reducir la identificacion directa del articulo. Los casos de una sola pagina o diseno no convencional deben revisarse manualmente antes de una publicacion abierta.

e la Médula Espinal; Compresión de la Médula Espinal; Artrodesis; Descompresión; Fracturas Óseas; Heridas

INTRODUCTION

Thoracic and lumbar fractures account for 30 - 50 % of all spinal injuries in trauma victims.[1] Patients with thoracic and lumbar spine fractures are mostly adults (mean age 39 years), economically active, and male.[2] Traumatic spinal fractures occur in about 6% of polytrauma patients, and about half (2.6%) have associated nerve injuries.[1]

To standardize and facilitate communication between doctors, the Frankel scale[3] describes neurological lesions. This is an alphabetical classification in which each letter represents a neurological loss. (Table 1)

When it comes to treatment, there are two approaches: conservative and surgical. The decision for which course of action to take should aim to restore spinal alignment, maintain stability and function, recover from neurological deficit and avoid its installation and progression, and resolve pain.[4]

An early approach (<48 hours) has shown the best results in the neurological recovery of patients with spinal cord injuries associated with neuromuscular deficits. Because spinal cord injury mainly affects the economically active population, it has a high individual social cost.[3]

This study aims to evaluate the neurological recovery at a 6-month follow-up in patients with thoracic and lumbar spine fractures who have undergone spinal decompression in less than 24 hours, between 24 and 48 hours, and more than 48 hours after the trauma.

METHODOLOGY

A retrospective evaluation of neurological recovery was performed by reviewing the medical records of patients with thoracic and lumbar fractures with neurological damage seen at an emergency hospital in Belo Horizonte, Minas Gerais, Brazil. It is a highly complex center financed by the Brazilian Unified Health System (SUS). (CAAE 01929818.0.0000.5119)

Data were collected between 2014 and 2018, and a database was built in Excel with the following variables: gender, age, fracture level, AO Spine classification, assessment of neurological status according to the Frankel scale (Initial and six months), and time to post-trauma surgery (< 24 hours, 24 - 48 hours or > 48 hours).

Following the hospital's flowchart, after clinical stability, the patients were evaluated by a spine surgeon and submitted to a CT scan. The patients were then classified neurologically by the Frankel scale, and the vertebral fractures were classified according to AO Spine.

The Frankel scale was introduced in 1969, allowing for a better understanding among hospital staff of the patient's neurological status. It is divided from A to E, "A" being the patient with a complete lesion, "B" with preserved sensitivity and absent motor, "C" preserved sensitivity and non-functional motor, "D" preserved sensitivity and functional motor, "E" the patient with no deficits. (Table 1)

After the surgical treatment, the patients were re-evaluated and reclassified for neurological recovery at return visits at 2, 6, 12, and 24 weeks according to the hospital's protocol.

Exclusion criteria were patients with cervical fractures, patients without neurological deficits, patients undergoing conservative

Table 1. Frankel Scale.

Frankel Scale		
Classification	**Motricity**	**Sensitivity**
A	Absent	Absent
B	Absent	Present
C	Present not useful	Present
D	Present useful	Present
E	Normal	Present

peripheral lesions, among others, and those patients who did not return for follow-up. In addition, patients with only thoracolumbar fractures, patients with neurological deficits, and those who attended return visits were included.

The study was approved by the ethics committee of João XXIII Hospital - FHEMIG -MG - Brazil.

Statistical analysis

The profile of the patients was characterized by absolute frequency (n), relative frequency (%) for categorical variables, and mean and standard deviation for continuous variables. The normality of the data was checked using the Shapiro-Wilk test. The comparison of neurological recovery, assessed according to Frankel's classification at baseline, six months, and one year after surgery as a function of the time of surgery, was performed using *Pearson*'s Chi-square test and the Kruskal-Wallis test. The Frankel scale was worked out statistically as an ordinal scale, ranging from 1 to 5, considering that the higher the grade, the better the patient's neurological status. The data were analyzed with the help of the *Statistical Package for Social Science*, version 26.0 (IBM Corporation, Armonk, USA). The significance level adopted was 5% ($p < 0.05$).

DISCUSSION

Spinal paralysis is responsible for considerable human suffering and an extraordinarily high hospital time and resource expenditure.[5]

SCI due to spinal fractures predominated in patients with a mean age of 34.20 ± 10.61 years, which coincides with the peak of the economically active population in Brazil, that is, 25 to 49 years-old,[6] which is why spinal fractures have a great economic impact. As for gender, a higher incidence was observed in male patients, which is in agreement with what was published by Lomaz and collaborators in 2017.[7]

The incidence of neurological involvement in patients with thoracic and lumbar spine injuries is variable.[8] Magerl et al.[9] found an overall incidence of 22%, while Frank et al.,[4] in a study of thoracic injuries only, found an 81.3% incidence of deficits. There is a great deal of variation among different studies on the neurological evolution of patients operated on for thoracic and lumbar spine injuries. Rath et al.,[8] reported a 71% rate of neurological improvement (at least 1st of Frankel). Our study obtained the highest improvement rate (36%) in patients operated on > 48 hours.

Rath et al.[8] in 2005 showed that even in cases of complete neurological injury (Frankel A), some patients could improve ent in the literature. Better neurologi- strated[6,11] in those patients operated . In contrast, McEvoy and Bradford[12] nt as performed within two weeks of injury. Experimental studies have shown a 6- to an 8-hour window of opportunity to significantly reduce secondary injury by relieving spinal cord pressure.[13,14]

To perform spine surgery in less than 24 hours, we encounter some difficulties, such as those patients who have accidents in neighboring cities and take time to be regulated and transferred to our hospital, as well as those who present with clinical instability that prevents them from having surgery immediately after arriving at our hospital.

As a limitation of the study, we emphasize that it is a retrospective study with a high number of patients with incomplete medical records or who did not return to our team and who were excluded from the study, thus determining a small sample for the study.

between patients who operated on early in less than 24 hours and those who operated on after more than 48 hours. The fact that the group of patients operated on > 48 hours is numerically larger may be an important confounding bias.

ACKNOWLEDGEMENTS

We thank our hospital staff, who made this study possible.

All authors declare no potential conflict of interest related to this article.

CONTRIBUTIONS OF THE AUTHORS: Each author contributed individually and significantly to the development of this article. SGM: design of the paper; data analysis and interpretation, writing of the manuscript; RLMR: a review of the intellectual concept of the paper and final approval of the manuscript version to be published; LFA: data analysis and revision of the paper; CAEM: substantial contribution in the design of the paper and critical review of its intellectual content.

REFERENCES

1. Bucholz RW, Heckman JD, Court-Brown, C, Tornetta III, P, Koval KJ, Wirth MA. Rockwood and Green's Fractures in Adults: Rockwood, Green, and Wilkins' Fractures. 8 ed. São Paulo: Wolters Kluwer; 2015.
2. Hu R, Mustard CA, Burns C. Epidemiology of incident spinal fracture in a complete population. Spine (Phila Pa 1976). 1996;21(4):492-9.
3. Frankel HL, Hancock DO, Hyslop G, Melzak J, Michaelis LS, Ungar GH, et al. The value of postural reduction in the initial management of closed injuries of the spine with paraplegia and tetraplegia. Paraplegia. 1969;7(3):179-92.
4. Boos N, Aebi M. Spinal Disorders, Fundamentals of Diagnosis and Treatment. Am J Neuroradiol. 2009;30(3):e44.
5. Bedbrook GM. The care and management of spinal cord injuries. Berling Heidelberg New York: Springer; 1981.
6. Instituto Brasileiro de Geografia e Estatística (IBGE). Indicadores IBGE: pesquisa mensal de emprego: abril 2015. Rio de Janeiro: IBGE; 2015. p. 1-29.
7. Lomaz MB, Netto LAFS, Filho MSG, Alves AP, Canto FRT. Perfil epidemiológico dos pacien- and its influencing factors in thoracic and lumbar spine fractures after surgical decompression and stabilization. Neurosurg Rev. 2005;28(1):44-52.
9. Magerl F, Aebi M, Gertzbein SD, Harms J, Nazarian AS. Comprehensive classification of thoracic and lumbar injuries. Eur Spine J. 1994;3(4):184-201.
10. Whitesides TE. Traumatic kyphosis of the thoracolumbar spine. Clin Orthop Relat Res. 1977;(128):78-92.
11. Wiber J, Hauge HN. Neurological outcome after surgery for thoracic and lumbar spine injuries. Acta Neurochir. 1988;91(3-4):106-12.
12. McEvoy RD, Bradford DS. The management of brust fractures of the thoracic and lumbar spine: experience in 53 patients. Spine. 1985;10(7):631-7.
13. Dilmar JR II, Glassman SD, Raque GH, Zhang YP, Shields CB. The influence of spinal canal narrowing and timing of descompression on neurologic recovery after spinal cord contusion in a rat model. Spine. 1999;24(16):1623-33.
14. Dolan EJ, Tator CH, Endrenyi L. The value of descompression for acute experimental spinal cord compression injury. J Neurosurg. 1980;53(6):749-55.